

## Unit 13. Matrix Differential Calculus I

# 1 First Differentials as Linear Maps

## 1.1 Differentiability and uniqueness

**Definition 13.1** (First differential). A set  $\mathcal{U} \subseteq \mathbb{R}^{m \times n}$  is **open** if, for every  $\mathbf{X} \in \mathcal{U}$ , some  $\rho > 0$  satisfies

$$\|\Delta\|_F < \rho \implies \mathbf{X} + \Delta \in \mathcal{U}.$$

Let  $\mathbf{F} : \mathcal{U} \rightarrow \mathbb{R}^{p \times q}$ . The function  $\mathbf{F}$  is **differentiable at  $\mathbf{X}$**  if a linear map  $\mathcal{L}_{\mathbf{X}} : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{p \times q}$  and a remainder  $\mathbf{R}_{\mathbf{X}}$  satisfy

$$\mathbf{F}(\mathbf{X} + \Delta) = \mathbf{F}(\mathbf{X}) + \mathcal{L}_{\mathbf{X}}(\Delta) + \mathbf{R}_{\mathbf{X}}(\Delta), \quad (1)$$

$$\forall \epsilon > 0 \exists \eta > 0 : \quad 0 < \|\Delta\|_F < \eta \implies \|\mathbf{R}_{\mathbf{X}}(\Delta)\|_F < \epsilon \|\Delta\|_F. \quad (2)$$

The **first differential** is

$$d\mathbf{F}_{\mathbf{X}}[\Delta] := \mathcal{L}_{\mathbf{X}}(\Delta).$$

For scalar output, replace the output Frobenius norm by absolute value. The notation  $\Delta \rightarrow \mathbf{O}$  means  $\|\Delta\|_F \rightarrow 0$ , and  $\mathbf{R}_{\mathbf{X}}(\Delta) = o(\|\Delta\|_F)$  abbreviates (2). The relation  $\mathbf{R}_{\mathbf{X}}(\Delta) = O(\|\Delta\|_F^k)$

means

$$\exists C, \eta > 0 : \quad \|\Delta\|_F < \eta \implies \|\mathbf{R}_{\mathbf{X}}(\Delta)\|_F \leq C \|\Delta\|_F^k.$$

For the identity map  $\text{id}(\mathbf{X}) = \mathbf{X}$ ,

$$\text{d id}_{\mathbf{X}}[\Delta] = \Delta;$$

thereafter  $\text{d}\mathbf{X}$  denotes the chosen increment  $\Delta$ .

Origin: Adapted from H §§15.1a, 15.1c, especially Eq. (15.1.1); A&M pp. 298–299; MC Eqs. (10)–(11)

**Lemma 13.2** (Uniqueness and continuity). The linear map in Definition 13.1 is unique. If  $\mathbf{F}$  is differentiable at  $\mathbf{X}$ , then

$$\mathbf{F}(\mathbf{X} + \Delta) \longrightarrow \mathbf{F}(\mathbf{X}) \quad (\Delta \rightarrow \mathbf{O}).$$

Origin: Course formulation from H Eq. (15.1.1); MC Eqs. (10)–(11)

**Proof.** Suppose  $\mathcal{L}_1, \mathcal{L}_2$  have remainders  $\mathbf{R}_1, \mathbf{R}_2$ . For fixed  $\mathbf{E} \neq \mathbf{O}$  and  $t \neq 0$ ,

$$(\mathcal{L}_1 - \mathcal{L}_2)(\mathbf{E}) = \frac{\mathbf{R}_2(t\mathbf{E}) - \mathbf{R}_1(t\mathbf{E})}{t} \longrightarrow \mathbf{O}.$$

Hence  $\mathcal{L}_1(\mathbf{E}) = \mathcal{L}_2(\mathbf{E})$ ; equality at  $\mathbf{E} = \mathbf{O}$  follows from linearity.

For the coordinate basis  $\{\mathbf{E}_{ij}\}$ , define  $\mathbf{M}_{\mathcal{L}}$  by

$$\mathbf{M}_{\mathcal{L}} \operatorname{vec} \mathbf{E}_{ij} = \operatorname{vec}\{\mathcal{L}(\mathbf{E}_{ij})\}.$$

Linearity and the  $\operatorname{vec}$  isometry give

$$\operatorname{vec}\{\mathcal{L}(\Delta)\} = \mathbf{M}_{\mathcal{L}} \operatorname{vec} \Delta, \quad \|\mathcal{L}(\Delta)\|_F \leq \|\mathbf{M}_{\mathcal{L}}\|_2 \|\Delta\|_F.$$

Thus both  $\mathcal{L}(\Delta)$  and  $\mathbf{R}_{\mathbf{X}}(\Delta)$  tend to zero in (1). □

## 1.2 Jacobians, gradients, and identification

**Definition 13.3** (Jacobian and matrix gradient). If  $\mathbf{F} : \mathcal{U} \rightarrow \mathbb{R}^{p \times q}$  is differentiable at  $\mathbf{X}$ , its **Jacobian** is the coordinate matrix of its differential:

$$\text{vec}\{\text{d}\mathbf{F}_{\mathbf{X}}[\Delta]\} = \mathbf{D}\mathbf{F}(\mathbf{X}) \text{vec } \Delta, \quad \mathbf{D}\mathbf{F}(\mathbf{X}) := \frac{\partial \text{vec } \mathbf{F}}{\partial (\text{vec } \mathbf{X})'}. \quad (3)$$

For differentiable  $f : \mathcal{U} \rightarrow \mathbb{R}$ , its **matrix gradient**  $\nabla_{\mathbf{X}} f \in \mathbb{R}^{m \times n}$  is defined by

$$\text{d}f_{\mathbf{X}}[\Delta] = \text{tr}\{(\nabla_{\mathbf{X}} f)' \Delta\}. \quad (4)$$

Origin: Adapted from A&M Eqs. (13.1)–(13.2); H §§15.1c, 15.1e–f; MC Eqs. (7)–(9)

### Coordinate consequences

If  $\mathbf{a}'$  is the coordinate row of the linear functional  $\text{d}f_{\mathbf{X}}$ , the vec bijection supplies the unique  $\mathbf{G}$  with  $\text{vec } \mathbf{G} = \mathbf{a}$ ; then  $\mathbf{a}' \text{vec } \Delta = \text{tr}(\mathbf{G}' \Delta)$ . Thus (4) always defines a matrix gradient. If  $\mathbf{E}_{ij}$  has

a 1 in position  $(i, j)$  and zeros elsewhere, then the Jacobian columns and gradient entries are

$$\mathbf{D}\mathbf{F}(\mathbf{X}) \operatorname{vec} \mathbf{E}_{ij} = \left. \frac{d}{dt} \operatorname{vec} \mathbf{F}(\mathbf{X} + t\mathbf{E}_{ij}) \right|_{t=0}, \quad (5)$$

$$(\nabla_{\mathbf{X}} f)_{ij} = \left. \frac{d}{dt} f(\mathbf{X} + t\mathbf{E}_{ij}) \right|_{t=0}. \quad (6)$$

To verify (5), substitute  $\Delta = t\mathbf{E}_{ij}$  in (1), divide by  $t$ , and let  $t \rightarrow 0$ . Equation (6) is its scalar coordinate form.

**Theorem 13.4** (First identification).

$$\left[ \operatorname{vec}\{d\mathbf{F}_{\mathbf{X}}[\Delta]\} = \mathbf{J} \operatorname{vec} \Delta \text{ for every } \Delta \right] \iff \mathbf{D}\mathbf{F}(\mathbf{X}) = \mathbf{J}, \quad (7)$$

$$\left[ df_{\mathbf{X}}[\Delta] = \operatorname{tr}(\mathbf{G}'\Delta) \text{ for every } \Delta \right] \iff \nabla_{\mathbf{X}} f = \mathbf{G}. \quad (8)$$

Consequently,

$$Df(\mathbf{X}) = \{\text{vec}(\nabla_{\mathbf{X}} f)\}' \quad (9)$$

Origin: Adapted from A&M Eq. (13.3) and Ex. 13.7; MC Prop. 11; H §15.6

**Proof.** The coordinate matrix of a linear map is unique, proving (7). By the vec inner-product identity,

$$\text{tr}(\mathbf{G}'\mathbf{\Delta}) = (\text{vec } \mathbf{G})' \text{vec } \mathbf{\Delta}.$$

Equality for every  $\mathbf{\Delta}$ , followed by  $\mathbf{\Delta} = \mathbf{E}_{ij}$ , proves (8). Equation (9) follows from the two identifications.  $\square$

## 1.3 Algebra and composition

**Theorem 13.5** (Algebra of first differentials). Let  $\mathbf{F}, \mathbf{G}$  be differentiable at  $\mathbf{X}$  and let  $\alpha, \beta$  be constants. With all products conformable,

$$d(\alpha \mathbf{F} + \beta \mathbf{G}) = \alpha d\mathbf{F} + \beta d\mathbf{G}, \quad (10)$$

$$d(\mathbf{F}') = (d\mathbf{F})', \quad d(\text{vec } \mathbf{F}) = \text{vec}(d\mathbf{F}), \quad (11)$$

$$d \text{tr } \mathbf{F} = \text{tr}(d\mathbf{F}), \quad (12)$$

$$d(\mathbf{F}\mathbf{G}) = (d\mathbf{F})\mathbf{G} + \mathbf{F}(d\mathbf{G}), \quad (13)$$

$$d(\mathbf{F} \otimes \mathbf{G}) = (d\mathbf{F}) \otimes \mathbf{G} + \mathbf{F} \otimes (d\mathbf{G}). \quad (14)$$

In (12),  $\mathbf{F}$  is square. All quantities on the right are evaluated at  $\mathbf{X}$  and in the same increment  $\Delta$ .

Origin: Adapted from A&M Exs. 13.1–13.3; H Lemmas 15.4.1–15.4.3 and Eqs. (15.4.9)–(15.4.10); MC Eqs. (12), (15)–(17)

**Proof.** Put  $h = \|\Delta\|_F$  and write

$$\mathbf{F}(\mathbf{X} + \Delta) = \mathbf{F} + \mathbf{L}_F + \mathbf{R}_F, \quad \mathbf{G}(\mathbf{X} + \Delta) = \mathbf{G} + \mathbf{L}_G + \mathbf{R}_G.$$

The coordinate-matrix bound in Lemma 13.2 gives

$$\|\mathbf{L}_F\|_F + \|\mathbf{L}_G\|_F = O(h), \quad \|\mathbf{R}_F\|_F + \|\mathbf{R}_G\|_F = o(h).$$

For the linear operations,

$$\mathbf{R}_{\alpha\mathbf{F}+\beta\mathbf{G}} = \alpha\mathbf{R}_F + \beta\mathbf{R}_G = o(h),$$

$$\|\mathbf{R}'_F\|_F = \|\mathbf{R}_F\|_F,$$

$$\|\text{vec } \mathbf{R}_F\|_2 = \|\mathbf{R}_F\|_F,$$

$$|\text{tr } \mathbf{R}_F| \leq \sqrt{p} \|\mathbf{R}_F\|_F \quad (\mathbf{F} \in \mathbb{R}^{p \times p}).$$

These bounds prove (10)–(12). For matrix multiplication, subtraction of its asserted linear part leaves

$$\mathbf{R}_F\mathbf{G} + \mathbf{F}\mathbf{R}_G + \mathbf{L}_F\mathbf{L}_G + \mathbf{L}_F\mathbf{R}_G + \mathbf{R}_F\mathbf{L}_G + \mathbf{R}_F\mathbf{R}_G.$$

The bounds

$$\|\mathbf{AB}\|_F \leq \|\mathbf{A}\|_2 \|\mathbf{B}\|_F, \quad \|\mathbf{A} \otimes \mathbf{B}\|_F = \|\mathbf{A}\|_F \|\mathbf{B}\|_F$$

make this sum  $o(h)$ . The identical six-term expansion with every product replaced by  $\otimes$  is  $o(h)$  by the second bound. This proves (13)–(14).  $\square$

**Theorem 13.6** (Chain rule). Let  $\mathbf{F} : \mathcal{U} \rightarrow \mathcal{V}$  be differentiable at  $\mathbf{X}$  and  $\mathbf{G} : \mathcal{V} \rightarrow \mathbb{R}^{r \times s}$  be differentiable at  $\mathbf{Y} = \mathbf{F}(\mathbf{X})$ , where  $\mathcal{U}, \mathcal{V}$  are open matrix sets. Then

$$d(\mathbf{G} \circ \mathbf{F})_{\mathbf{X}}[\Delta] = d\mathbf{G}_{\mathbf{Y}}[d\mathbf{F}_{\mathbf{X}}[\Delta]], \quad (15)$$

$$D(\mathbf{G} \circ \mathbf{F})(\mathbf{X}) = D\mathbf{G}(\mathbf{Y})D\mathbf{F}(\mathbf{X}). \quad (16)$$

Origin: Adapted from H Thm. 15.7.1 and Eqs. (15.7.7)–(15.7.10); A&M pp. 299–300; MC Prop. 12

**Proof.** Put

$$\mathbf{K} = \mathbf{F}(\mathbf{X} + \Delta) - \mathbf{F}(\mathbf{X}) = \mathcal{L}_F(\Delta) + \mathbf{R}_F(\Delta).$$

Lemma 13.2 gives  $\mathbf{K} \rightarrow \mathbf{O}$  and  $\|\mathbf{K}\|_F = O(\|\Delta\|_F)$ . Hence

$$\mathbf{G}(\mathbf{Y} + \mathbf{K}) - \mathbf{G}(\mathbf{Y}) = \mathcal{L}_G \mathcal{L}_F(\Delta) + \mathcal{L}_G \mathbf{R}_F(\Delta) + \mathbf{R}_G(\mathbf{K}),$$

$$\frac{\|\mathcal{L}_G \mathbf{R}_F(\Delta)\|_F}{\|\Delta\|_F} \longrightarrow 0,$$

$$\frac{\|\mathbf{R}_G(\mathbf{K})\|_F}{\|\Delta\|_F} = \frac{\|\mathbf{R}_G(\mathbf{K})\|_F}{\|\mathbf{K}\|_F} \frac{\|\mathbf{K}\|_F}{\|\Delta\|_F} \longrightarrow 0.$$

If  $\mathbf{K} = \mathbf{O}$ , take the first factor in the last line as 0. Thus the linear part is  $\mathcal{L}_G \mathcal{L}_F$ . Coordinate matrices multiply, giving (16).  $\square$

## 2 Reusable Matrix Differentials

### 2.1 Trace forms and powers

**Theorem 13.7** (Trace-gradient formulas). Let  $\mathbf{X} \in \mathbb{R}^{m \times n}$  be unrestricted. All constants are conformable; in the quadratic trace formula,  $\mathbf{A}$  and  $\mathbf{B}$  have orders  $m$  and  $n$ .

$$f(\mathbf{X}) = \text{tr}(\mathbf{A}'\mathbf{X}) : \quad df = \text{tr}(\mathbf{A}'d\mathbf{X}), \quad \nabla_{\mathbf{X}}f = \mathbf{A}, \quad (17)$$

$$f(\mathbf{X}) = \mathbf{a}'\mathbf{X}\mathbf{b} : \quad df = \mathbf{a}'(d\mathbf{X})\mathbf{b}, \quad \nabla_{\mathbf{X}}f = \mathbf{a}\mathbf{b}', \quad (18)$$

$$f(\mathbf{X}) = \text{tr}(\mathbf{X}'\mathbf{A}\mathbf{X}\mathbf{B}) : \quad \nabla_{\mathbf{X}}f = \mathbf{A}\mathbf{X}\mathbf{B} + \mathbf{A}'\mathbf{X}\mathbf{B}', \quad (19)$$

$$f(\mathbf{X}) = \|\mathbf{A}\mathbf{X}\mathbf{B} - \mathbf{C}\|_F^2 : \quad \nabla_{\mathbf{X}}f = 2\mathbf{A}'(\mathbf{A}\mathbf{X}\mathbf{B} - \mathbf{C})\mathbf{B}'. \quad (20)$$

Origin: Adapted from H §15.6 and Exs. 15.8, 15.11; A&M Exs. 13.6–13.7, 13.10; MC Ex. 19

**Proof.** The differential rules and trace cyclicity give

$$\begin{aligned} d \operatorname{tr}(\mathbf{A}'\mathbf{X}) &= \operatorname{tr}(\mathbf{A}'d\mathbf{X}), \\ d(\mathbf{a}'\mathbf{X}\mathbf{b}) &= \mathbf{a}'(d\mathbf{X})\mathbf{b} = \operatorname{tr}\{(\mathbf{a}\mathbf{b}')'d\mathbf{X}\}, \\ d \operatorname{tr}(\mathbf{X}'\mathbf{A}\mathbf{X}\mathbf{B}) &= \operatorname{tr}\{(d\mathbf{X})'\mathbf{A}\mathbf{X}\mathbf{B} + \mathbf{X}'\mathbf{A}(d\mathbf{X})\mathbf{B}\} \\ &= \operatorname{tr}\{(\mathbf{A}\mathbf{X}\mathbf{B} + \mathbf{A}'\mathbf{X}\mathbf{B}')'d\mathbf{X}\}. \end{aligned}$$

For  $\mathbf{R} = \mathbf{A}\mathbf{X}\mathbf{B} - \mathbf{C}$ ,

$$d\|\mathbf{R}\|_F^2 = d \operatorname{tr}(\mathbf{R}'\mathbf{R}) = 2 \operatorname{tr}\{\mathbf{R}'\mathbf{A}(d\mathbf{X})\mathbf{B}\} = 2 \operatorname{tr}\{(\mathbf{A}'\mathbf{R}\mathbf{B}')'d\mathbf{X}\}.$$

Theorem 13.4 identifies all four gradients. □

**Theorem 13.8** (Matrix and trace powers). For square  $\mathbf{X}$  and integer  $k \geq 1$ ,

$$d(\mathbf{X}^k) = \sum_{j=0}^{k-1} \mathbf{X}^j (d\mathbf{X}) \mathbf{X}^{k-1-j}, \quad (21)$$

$$D(\mathbf{X}^k) = \sum_{j=0}^{k-1} (\mathbf{X}^{k-1-j})' \otimes \mathbf{X}^j, \quad (22)$$

$$d \operatorname{tr}(\mathbf{X}^k) = k \operatorname{tr}\{\mathbf{X}^{k-1} d\mathbf{X}\}, \quad \nabla_{\mathbf{X}} \operatorname{tr}(\mathbf{X}^k) = k(\mathbf{X}^{k-1})'. \quad (23)$$

Origin: Adapted from A&M Exs. 13.8, 13.18; H Exs. 15.10, 16.31; S Ex. 10.24; MC Ex. 21

**Proof.** The product rule proves (21) by induction. Roth's identity applied termwise gives (22). Cyclicity gives

$$\sum_{j=0}^{k-1} \operatorname{tr}\{\mathbf{X}^j (d\mathbf{X}) \mathbf{X}^{k-1-j}\} = \sum_{j=0}^{k-1} \operatorname{tr}\{\mathbf{X}^{k-1} d\mathbf{X}\}.$$

The gradient follows by identification. □

## 2.2 Inverse, determinant, and log-determinant

**Theorem 13.9** (Inverse and linear-solve sensitivity). For nonsingular  $\mathbf{X} \in \mathbb{R}^{n \times n}$ ,

$$d(\mathbf{X}^{-1}) = -\mathbf{X}^{-1}(d\mathbf{X})\mathbf{X}^{-1}, \quad (24)$$

$$D(\mathbf{X}^{-1}) = -\mathbf{X}^{-T} \otimes \mathbf{X}^{-1}. \quad (25)$$

If  $\mathbf{A}(\mathbf{Z})$  is nonsingular and  $\mathbf{A}(\mathbf{Z}), \mathbf{B}(\mathbf{Z})$  are differentiable, then  $\mathbf{Y} = \mathbf{A}^{-1}\mathbf{B}$  satisfies

$$d\mathbf{Y} = \mathbf{A}^{-1}\{d\mathbf{B} - (d\mathbf{A})\mathbf{Y}\}. \quad (26)$$

Origin: Adapted from H Eqs. (15.8.14)–(15.8.18), (16.6.13), (16.6.15); A&M Exs. 13.21, 13.25; MC Eq. (32) and Ex. 22

**Proof.** Put  $\mathbf{E} = \mathbf{X}^{-1}\Delta$ . If  $\|\mathbf{E}\|_2 < 1$ , then

$$\|(\mathbf{I} + \mathbf{E})\mathbf{z}\|_2 \geq (1 - \|\mathbf{E}\|_2)\|\mathbf{z}\|_2.$$

Thus  $\text{null}(\mathbf{I} + \mathbf{E}) = \{\mathbf{0}\}$ , so  $\mathbf{I} + \mathbf{E}$  is nonsingular. For arbitrary  $\mathbf{y}$ , take  $\mathbf{z} = (\mathbf{I} + \mathbf{E})^{-1}\mathbf{y}$  to obtain

$$\|(\mathbf{I} + \mathbf{E})^{-1}\|_2 \leq \frac{1}{1 - \|\mathbf{E}\|_2}.$$

Moreover,

$$(\mathbf{I} + \mathbf{E})^{-1} = \mathbf{I} - \mathbf{E} + \mathbf{E}^2(\mathbf{I} + \mathbf{E})^{-1}.$$

Therefore

$$\begin{aligned} (\mathbf{X} + \mathbf{\Delta})^{-1} &= \mathbf{X}^{-1} - \mathbf{X}^{-1}\mathbf{\Delta}\mathbf{X}^{-1} \\ &\quad + (\mathbf{X}^{-1}\mathbf{\Delta})^2(\mathbf{I} + \mathbf{X}^{-1}\mathbf{\Delta})^{-1}\mathbf{X}^{-1}, \end{aligned}$$

and the last term is  $O(\|\mathbf{\Delta}\|_F^2)$ . This proves (24); Roth's identity gives (25). Finally,

$$d(\mathbf{A}^{-1}\mathbf{B}) = -\mathbf{A}^{-1}(d\mathbf{A})\mathbf{A}^{-1}\mathbf{B} + \mathbf{A}^{-1}d\mathbf{B},$$

which is (26). □

**Theorem 13.10** (Determinant and log-determinant). For every square  $\mathbf{X}$ ,

$$d \det \mathbf{X} = \operatorname{tr}\{\operatorname{adj}(\mathbf{X})d\mathbf{X}\}, \quad \nabla_{\mathbf{X}} \det \mathbf{X} = \operatorname{adj}(\mathbf{X})'. \quad (27)$$

If  $\mathbf{X}$  is nonsingular, then

$$d \det \mathbf{X} = (\det \mathbf{X}) \operatorname{tr}(\mathbf{X}^{-1}d\mathbf{X}), \quad \nabla_{\mathbf{X}} \det \mathbf{X} = (\det \mathbf{X})\mathbf{X}^{-T}, \quad (28)$$

$$d \log|\det \mathbf{X}| = \operatorname{tr}(\mathbf{X}^{-1}d\mathbf{X}), \quad \nabla_{\mathbf{X}} \log|\det \mathbf{X}| = \mathbf{X}^{-T}. \quad (29)$$

For positive definite  $\mathbf{X}$ ,  $\log|\det \mathbf{X}| = \log \det \mathbf{X}$ .

Origin: Adapted from H Eqs. (15.8.1)–(15.8.8); A&M Exs. 13.32, 13.36; MC Eqs. (30)–(31)

**Proof.** Write  $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_n)$  and  $\Delta = (\delta_1, \dots, \delta_n)$ . Multilinearity gives

$$\det(\mathbf{X} + \Delta) = \det \mathbf{X} + \sum_{j=1}^n \det(\mathbf{x}_1, \dots, \delta_j, \dots, \mathbf{x}_n) + O(\|\Delta\|_F^2).$$

Laplace expansion identifies the linear sum as  $\sum_{ij} c_{ij}\delta_{ij} = \operatorname{tr}\{\operatorname{adj}(\mathbf{X})\Delta\}$ , where  $c_{ij}$  is the  $(i, j)$

cofactor. This proves (27). If  $\mathbf{X}$  is nonsingular,  $\text{adj}(\mathbf{X}) = (\det \mathbf{X})\mathbf{X}^{-1}$ , giving (28). The one-variable chain rule and  $(d/ds) \log|s| = s^{-1}$  for  $s \neq 0$  give (29).  $\square$

# 3 Jacobians and Structural Sensitivity

## 3.1 Coordinate Jacobians

**Theorem 13.11** (Three coordinate Jacobians). Let  $\mathbf{X} \in \mathbb{R}^{m \times n}$ .

$$\mathbf{F}(\mathbf{X}) = \mathbf{A}\mathbf{X}\mathbf{B} + \mathbf{C} : \quad \mathbf{D}\mathbf{F}(\mathbf{X}) = \mathbf{B}' \otimes \mathbf{A}, \quad (30)$$

$$\mathbf{T}(\mathbf{X}) = \mathbf{X}' : \quad \mathbf{D}\mathbf{T}(\mathbf{X}) = \mathbf{K}_{mn}, \quad (31)$$

$$\mathbf{G}(\mathbf{X}) = \mathbf{X}'\mathbf{X} : \quad \mathbf{D}\mathbf{G}(\mathbf{X}) = (\mathbf{X}' \otimes \mathbf{I}_n)\mathbf{K}_{mn} + \mathbf{I}_n \otimes \mathbf{X}'. \quad (32)$$

In (30),  $\mathbf{A} \in \mathbb{R}^{r \times m}$  and  $\mathbf{B} \in \mathbb{R}^{n \times s}$ , and  $\mathbf{C} \in \mathbb{R}^{r \times s}$ .

Origin: Adapted from A&M Exs. 13.17, 13.19; H Eqs. (15.4.3), (15.4.7), (15.4.10), (15.5.3), (16.6.6), (16.6.8); MC §§4.4–4.5

**Proof.** The differentials are

$$d\mathbf{F} = \mathbf{A}(d\mathbf{X})\mathbf{B}, \quad d\mathbf{T} = (d\mathbf{X})', \quad d\mathbf{G} = (d\mathbf{X})'\mathbf{X} + \mathbf{X}'d\mathbf{X}.$$

Roth's identity and  $\text{vec}(d\mathbf{X})' = \mathbf{K}_{mn} \text{vec } d\mathbf{X}$  give the three coordinate matrices. □

**Theorem 13.12** (Jacobian determinants of two transformations). Let  $\mathbf{A} \in \mathbb{R}^{m \times m}$  and  $\mathbf{B} \in \mathbb{R}^{n \times n}$  be nonsingular, and let  $\mathbf{X} \in \mathbb{R}^{m \times n}$ .

$$\mathbf{Y} = \mathbf{A}\mathbf{X}\mathbf{B} : \quad \det \left\{ \frac{\partial \text{vec } \mathbf{Y}}{\partial (\text{vec } \mathbf{X})'} \right\} = (\det \mathbf{A})^n (\det \mathbf{B})^m, \quad (33)$$

$$\mathbf{Y} = \mathbf{X}^{-1}, \quad \mathbf{X} \in \mathbb{R}^{n \times n} \text{ nonsingular} : \quad \det \left\{ \frac{\partial \text{vec } \mathbf{Y}}{\partial (\text{vec } \mathbf{X})'} \right\} = (-1)^n (\det \mathbf{X})^{-2n}. \quad (34)$$

For nonsingular  $\mathbf{X} = \mathbf{X}'$  and vech input and output coordinates,

$$\det \left\{ \frac{\partial \text{vech}(\mathbf{X}^{-1})}{\partial (\text{vech } \mathbf{X})'} \right\} = (-1)^{n(n+1)/2} (\det \mathbf{X})^{-(n+1)}. \quad (35)$$

Origin: Adapted from A&M Exs. 13.37–13.38; H Eqs. (16.6.9), (16.6.16), (16.6.20); MC §7

**Proof.** Equation (30) and the Kronecker determinant give (33). From (25),

$$\begin{aligned}\det(-\mathbf{X}^{-T} \otimes \mathbf{X}^{-1}) &= (-1)^{n^2} \det(\mathbf{X}^{-T})^n \det(\mathbf{X}^{-1})^n \\ &= (-1)^n (\det \mathbf{X})^{-2n}.\end{aligned}$$

For symmetric  $\mathbf{X}$ ,

$$d \operatorname{vech}(\mathbf{X}^{-1}) = -\mathbf{C}_{\mathbf{X}^{-1}} d \operatorname{vech} \mathbf{X}.$$

The dimension is  $n(n+1)/2$ , and  $\det \mathbf{C}_{\mathbf{X}^{-1}} = (\det \mathbf{X})^{-(n+1)}$  by the earlier symmetric-congruence coordinate determinant.  $\square$

## 3.2 Projection and eigenpair sensitivity

**Theorem 13.13** (Differential of a full-rank projector). Let  $\mathbf{X} \in \mathbb{R}^{n \times p}$  have full column rank and put

$$\mathbf{R} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}', \quad \mathbf{P} = \mathbf{X}\mathbf{R}, \quad \mathbf{M} = \mathbf{I}_n - \mathbf{P}.$$

Then

$$d\mathbf{P} = \mathbf{M}(d\mathbf{X})\mathbf{R} + \mathbf{R}'(d\mathbf{X})'\mathbf{M}. \quad (36)$$

Origin: Adapted from H Thm. 15.11.1, Eq. (15.11.1) with  $\mathbf{W} = \mathbf{I}$ ; A&M Ex. 13.24

**Proof.** Let  $\mathbf{S} = (\mathbf{X}'\mathbf{X})^{-1}$ . The product and inverse rules give

$$\begin{aligned} d\mathbf{P} &= (d\mathbf{X})\mathbf{S}\mathbf{X}' + \mathbf{X}\mathbf{S}(d\mathbf{X})' \\ &\quad - \mathbf{X}\mathbf{S}\{(\mathbf{X}')'\mathbf{X} + \mathbf{X}'(d\mathbf{X})\}\mathbf{S}\mathbf{X}' \\ &= (\mathbf{I} - \mathbf{P})(d\mathbf{X})\mathbf{S}\mathbf{X}' + \mathbf{X}\mathbf{S}(d\mathbf{X})'(\mathbf{I} - \mathbf{P}), \end{aligned}$$

which is (36) because  $\mathbf{R} = \mathbf{S}\mathbf{X}'$  and  $\mathbf{R}' = \mathbf{X}\mathbf{S}$ . □

**Theorem 13.14** (Derivative along a differentiable symmetric eigenpair). Assume that  $\mathbf{A}(t) = \mathbf{A}(t)'$  and a normalized eigenpair branch  $(\lambda_i(t), \mathbf{q}_i(t))$  are differentiable at  $t = 0$  and satisfy

$$\mathbf{A}(t)\mathbf{q}_i(t) = \lambda_i(t)\mathbf{q}_i(t), \quad \mathbf{q}_i(t)'\mathbf{q}_i(t) = 1.$$

Write a dot for  $d/dt$  at 0. Then

$$\dot{\lambda}_i = \mathbf{q}_i' \dot{\mathbf{A}} \mathbf{q}_i, \quad (37)$$

$$\mathbf{q}_i' \dot{\mathbf{q}}_i = 0. \quad (38)$$

If  $\lambda_i$  is simple and  $\mathbf{A}\mathbf{q}_j = \lambda_j\mathbf{q}_j$  for an orthonormal eigenbasis, then

$$\dot{\mathbf{q}}_i = \sum_{j \neq i} \frac{\mathbf{q}_j' \dot{\mathbf{A}} \mathbf{q}_i}{\lambda_i - \lambda_j} \mathbf{q}_j. \quad (39)$$

Origin: Adapted from H Eqs. (21.15.1)–(21.15.3), under the simplicity hypothesis of Lemma 21.15.1

**Proof.** Differentiate the eigen-equation and the normalization:

$$\begin{aligned}(\mathbf{A} - \lambda_i \mathbf{I})\dot{\mathbf{q}}_i &= -(\dot{\mathbf{A}} - \dot{\lambda}_i \mathbf{I})\mathbf{q}_i, \\ 2\mathbf{q}'_i \dot{\mathbf{q}}_i &= 0.\end{aligned}$$

Premultiplication of the first equation by  $\mathbf{q}'_i$  gives (37). For  $j \neq i$ ,

$$(\lambda_j - \lambda_i)\mathbf{q}'_j \dot{\mathbf{q}}_i = -\mathbf{q}'_j \dot{\mathbf{A}} \mathbf{q}_i.$$

Expand  $\dot{\mathbf{q}}_i$  in the orthonormal eigenbasis and use (38). □

**Definition 13.15** (Stationary point). Let  $f : \mathcal{U} \rightarrow \mathbb{R}$  be differentiable on an open matrix set. An interior point  $\mathbf{X} \in \mathcal{U}$  is **stationary** if

$$\mathrm{d}f_{\mathbf{X}}[\Delta] = 0 \quad \text{for every } \Delta. \tag{40}$$

By Theorem 13.4,

$$\mathbf{X} \text{ is stationary} \iff \nabla_{\mathbf{X}} f = \mathbf{0} \iff Df(\mathbf{X}) = \mathbf{0}'. \quad (41)$$

Origin: Adapted from A&M p. 300; MC §5, opening paragraph

## 4 Computation: Numerical Checks in R

### 4.1 A quadratic matrix map (§1.1)

For  $\mathbf{F}(\mathbf{X}) = \mathbf{X}'\mathbf{X}$ ,

$$\mathbf{F}(\mathbf{X} + h\mathbf{E}) - \mathbf{F}(\mathbf{X}) - h\{\mathbf{E}'\mathbf{X} + \mathbf{X}'\mathbf{E}\} = h^2\mathbf{E}'\mathbf{E}.$$

```

1 set.seed(1301)
2 X <- matrix(rnorm(15), 5, 3)
3 E <- matrix(rnorm(15), 5, 3)
4 dF <- t(E) %*% X + t(X) %*% E
5 h <- 10^-(1:8)
6 err <- sapply(h, function(step) {
7   R <- t(X + step * E) %*% (X + step * E) - t(X) %*% X - step * dF
8   sqrt(sum(R^2))

```

```

9  })
10 cbind(h = h, error_over_h = err / h, error_over_h2 = err / h^2)

```

```

##           h  error_over_h error_over_h2
## [1,] 1e-01 0.95383045674          9.538
## [2,] 1e-02 0.09538304567          9.538
## [3,] 1e-03 0.00953830457          9.538
## [4,] 1e-04 0.00095383046          9.538
## [5,] 1e-05 0.00009538306          9.538
## [6,] 1e-06 0.00000953776          9.538
## [7,] 1e-07 0.00000095472          9.547
## [8,] 1e-08 0.00000006861          6.861

```

The printed `error_over_h` tends to zero; `error_over_h2` is constant up to rounding.

## 4.2 Jacobian of a two-sided linear map (§3.1)

```

1  set.seed(1302)
2  m <- 2; n <- 3; r <- 4; s <- 2
3  A <- matrix(rnorm(r * m), r, m)
4  B <- matrix(rnorm(n * s), n, s)
5  X <- matrix(rnorm(m * n), m, n)

```

```
6 map <- function(z) c(A %*% matrix(z, m, n) %*% B)
7 step <- 1e-6
8 numeric_J <- sapply(seq_len(m * n), function(j) {
9   ej <- numeric(m * n); ej[j] <- 1
10   (map(c(X) + step * ej) - map(c(X) - step * ej)) / (2 * step)
11 })
12 analytic_J <- t(B) %x% A
13 c(maximum_entry_error = max(abs(numeric_J - analytic_J)),
14   tolerance = 5e-9)
```

```
## maximum_entry_error      tolerance
##           5.702e-10         5.000e-09
```

## 4.3 Inverse and log-determinant (§2.2)

```
1 set.seed(1303)
2 Z <- matrix(rnorm(16), 4, 4)
3 X <- crossprod(Z) + diag(4)
4 E <- matrix(rnorm(16), 4, 4)
5 Xinv <- solve(X)
6 d_inverse <- -Xinv %*% E %*% Xinv
7 d_logdet <- sum(diag(Xinv %*% E))
```

```

8 logabsdet <- function(M) as.numeric(determinant(M, logarithm = TRUE)$modulus)
9 h <- 10^-(1:7)
10 inverse_error <- sapply(h, function(step)
11   sqrt(sum((solve(X + step * E) - Xinv - step * d_inverse)^2)))
12 logdet_error <- sapply(h, function(step)
13   abs(logabsdet(X + step * E) - logabsdet(X) - step * d_logdet))
14 cbind(h = h,
15       inverse_error_over_h2 = inverse_error / h^2,
16       logdet_error_over_h2 = logdet_error / h^2)

```

| ##      | h         | inverse_error_over_h2 | logdet_error_over_h2 |
|---------|-----------|-----------------------|----------------------|
| ## [1,] | 0.1000000 | 0.5037                | 0.02753              |
| ## [2,] | 0.0100000 | 0.5083                | 0.02957              |
| ## [3,] | 0.0010000 | 0.5088                | 0.02978              |
| ## [4,] | 0.0001000 | 0.5088                | 0.02980              |
| ## [5,] | 0.0000100 | 0.5088                | 0.02981              |
| ## [6,] | 0.0000010 | 0.5087                | 0.03059              |
| ## [7,] | 0.0000001 | 0.5129                | 0.08384              |

The two scaled errors remain bounded until floating-point cancellation dominates.

## 4.4 Projection sensitivity (§3.2)

```

1  set.seed(1304)
2  X <- matrix(rnorm(21), 7, 3)
3  E <- matrix(rnorm(21), 7, 3)
4  projector <- function(Z) Z %*% solve(t(Z) %*% Z, t(Z))
5  P <- projector(X); M <- diag(7) - P
6  R <- solve(t(X) %*% X, t(X))
7  dP <- M %*% E %*% R + t(R) %*% t(E) %*% M
8  h <- 10^-(1:7)
9  err <- sapply(h, function(step)
10     sqrt(sum((projector(X + step * E) - P - step * dP)^2)))
11  c(tangent_block = max(abs(P %*% dP %*% P)),
12     symmetry_error = max(abs(dP - t(dP))),
13     max_scaled_remainder = max(err[1:5] / h[1:5]^2),
14     structural_zero_tolerance = 2e-10)

```

|    |                      |                           |
|----|----------------------|---------------------------|
| ## | tangent_block        | symmetry_error            |
| ## | 5.878e-17            | 1.110e-16                 |
| ## | max_scaled_remainder | structural_zero_tolerance |
| ## | 1.416e+00            | 2.000e-10                 |

## 4.5 Simple and repeated eigenvalues (§3.2)

```

1  set.seed(1305)
2  Q <- qr.Q(qr(matrix(rnorm(16), 4, 4)))
3  A <- Q %*% diag(c(1, 2.5, 5, 8)) %*% t(Q)
4  E0 <- matrix(rnorm(16), 4, 4); E <- (E0 + t(E0)) / 2
5  q <- Q[, 4]
6  analytic <- drop(t(q) %*% E %*% q)
7  lambda_max <- function(M) max(eigen(M, symmetric = TRUE, only.values = TRUE)$values)
8  h <- 10^-(1:7)
9  simple_error <- sapply(h, function(step)
10     abs((lambda_max(A + step * E) - 8) / step - analytic))
11  A_rep <- diag(c(3, 3, 1)); E_rep <- diag(c(1, -1, 0))
12  side_steps <- c(-1e-6, 1e-6)
13  side_quotients <- sapply(side_steps, function(step)
14     (lambda_max(A_rep + step * E_rep) - 3) / step)
15  list(simple_branch = cbind(h = h, derivative_error = simple_error),
16       repeated_branch = cbind(h = side_steps, quotient = side_quotients))

```

```
## $simple_branch
##           h derivative_error
```

```
## [1,] 0.1000000    0.03447036038
## [2,] 0.0100000    0.00326435675
## [3,] 0.0010000    0.00032474620
## [4,] 0.0001000    0.00003245786
## [5,] 0.0000100    0.00000324563
## [6,] 0.0000010    0.00000032832
## [7,] 0.0000001    0.00000004765
##
## $repeated_branch
##           h quotient
## [1,] -0.000001      -1
## [2,]  0.000001       1
```

The simple-branch errors tend to zero. At the repeated eigenvalue, the two one-sided quotients are  $-1$  and  $1$ .

## 5 Pitfalls

### 5.1 Using a matrix-shaped partial array as a Jacobian

For  $\mathbf{X} \in \mathbb{R}^{n \times n}$  and  $f(\mathbf{X}) = \text{tr } \mathbf{X}$ ,

$$\nabla_{\mathbf{X}} f = \mathbf{I}_n, \quad \mathbf{D}f(\mathbf{X}) = (\text{vec } \mathbf{I}_n)'.$$

The first object has order  $n \times n$ ; the Jacobian has order  $1 \times n^2$ .

### 5.2 Commuting an increment through a matrix

$$\mathrm{d}(\mathbf{X}^2) = (\mathrm{d}\mathbf{X})\mathbf{X} + \mathbf{X}(\mathrm{d}\mathbf{X}),$$

which equals  $2\mathbf{X}\mathrm{d}\mathbf{X}$  only when  $\mathbf{X}\mathrm{d}\mathbf{X} = (\mathrm{d}\mathbf{X})\mathbf{X}$ .

### 5.3 Using inverse formulas at a singular matrix

The identity

$$\mathrm{d} \det \mathbf{X} = \text{tr}\{\text{adj}(\mathbf{X})\mathrm{d}\mathbf{X}\}$$

holds at singular matrices. The formulas containing  $\mathbf{X}^{-1}$  and  $\log|\det \mathbf{X}|$  require  $\det \mathbf{X} \neq 0$ .

## 5.4 Treating symmetric coordinates as independent vec coordinates

If  $\mathbf{X} = \mathbf{X}'$ , then

$$d \operatorname{vec} \mathbf{X} = \mathbf{D}_n d \operatorname{vech} \mathbf{X}.$$

Consequently an unrestricted Jacobian  $\mathbf{J}$  becomes  $\mathbf{J}\mathbf{D}_n$  on the input side; a symmetric output requires  $\mathbf{D}_r^+$  on the output side.

## 5.5 Applying a simple-eigenvalue formula at a repeated eigenvalue

Equation (39) contains  $(\lambda_i - \lambda_j)^{-1}$ . If  $\lambda_i = \lambda_j$ , the eigenvector is not unique, and an ordered eigenvalue may have different one-sided derivatives.

## 6 Exercises

**Exercise 13.1** (Partials without a differential). Define

$$f(x, y) = \begin{cases} \frac{xy}{\sqrt{x^2 + y^2}}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Prove that both partial derivatives at  $(0, 0)$  exist, but  $f$  is not differentiable there.

Origin: Course formulation from H §15.1c and A&M pp. 298–299

**Solution.** On the coordinate axes,  $f(t, 0) = f(0, t) = 0$ , so both partial derivatives are 0. A differential would therefore have linear part  $L(h, k) = 0$ . For  $t \neq 0$ ,

$$\frac{|f(t, t) - L(t, t)|}{\|(t, t)'\|_2} = \frac{|t|/\sqrt{2}}{\sqrt{2}|t|} = \frac{1}{2} \not\rightarrow 0.$$



**Exercise 13.2** (Normalizing a vector). For  $\mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ , put  $\mathbf{u} = \mathbf{x}/\|\mathbf{x}\|_2$ . Prove

$$d\|\mathbf{x}\|_2 = \mathbf{u}'d\mathbf{x}, \quad d\mathbf{u} = \frac{1}{\|\mathbf{x}\|_2}(\mathbf{I} - \mathbf{u}\mathbf{u}')d\mathbf{x}.$$

Identify  $D\mathbf{u}(\mathbf{x})$  and its null space.

Origin: Course formulation from A&M Ex. 13.5 and H Thm. 15.7.1

**Solution.** For  $r = (\mathbf{x}'\mathbf{x})^{1/2} > 0$ , the scalar square-root derivative and Theorem 13.6 give

$$dr = \frac{1}{2r}d(\mathbf{x}'\mathbf{x}) = \frac{\mathbf{x}'d\mathbf{x}}{r} = \mathbf{u}'d\mathbf{x}.$$

Hence

$$d\mathbf{u} = r^{-1}d\mathbf{x} - r^{-2}\mathbf{x}dr = r^{-1}(\mathbf{I} - \mathbf{u}\mathbf{u}')d\mathbf{x}.$$

Thus  $D\mathbf{u} = r^{-1}(\mathbf{I} - \mathbf{u}\mathbf{u}')$  and  $\text{null}(D\mathbf{u}) = \text{span}\{\mathbf{x}\}$ . ■

**Exercise 13.3** (Generalized Rayleigh quotient). Let  $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$ , with  $\mathbf{A} = \mathbf{A}'$  and  $\mathbf{B} = \mathbf{B}' \succ \mathbf{O}$ . For  $\mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ , put

$$r(\mathbf{x}) = \frac{\mathbf{x}'\mathbf{A}\mathbf{x}}{\mathbf{x}'\mathbf{B}\mathbf{x}}.$$

Prove

$$\nabla_{\mathbf{x}} r = \frac{2}{\mathbf{x}'\mathbf{B}\mathbf{x}} \{\mathbf{A}\mathbf{x} - r(\mathbf{x})\mathbf{B}\mathbf{x}\}.$$

Origin: Adapted from H Eqs. (15.2.6), (15.3.7); A&M Ex. 13.4(b)

**Solution.** Let  $a = \mathbf{x}'\mathbf{A}\mathbf{x}$  and  $b = \mathbf{x}'\mathbf{B}\mathbf{x} > 0$ . Then

$$\begin{aligned} dr &= b^{-1}da - ab^{-2}db \\ &= \frac{2}{b} \{\mathbf{A}\mathbf{x} - (a/b)\mathbf{B}\mathbf{x}\}' d\mathbf{x}. \end{aligned}$$

Identification gives the result. 

**Exercise 13.4** (Jacobian of a Kronecker product). Let  $\mathbf{X} \in \mathbb{R}^{p \times q}$ ,  $\mathbf{Y} \in \mathbb{R}^{r \times s}$ ,

$$\mathbf{z} = \begin{pmatrix} \text{vec } \mathbf{X} \\ \text{vec } \mathbf{Y} \end{pmatrix}, \quad \mathbf{P} = \mathbf{I}_q \otimes \mathbf{K}_{sp} \otimes \mathbf{I}_r, \quad \mathbf{H}(\mathbf{z}) = \mathbf{X} \otimes \mathbf{Y}.$$

Prove

$$\begin{aligned} \text{vec}(\mathbf{X} \otimes \mathbf{Y}) &= \mathbf{P}(\text{vec } \mathbf{X} \otimes \text{vec } \mathbf{Y}), \\ \mathbf{DH}(\mathbf{z}) &= \mathbf{P}(\mathbf{I}_{pq} \otimes \text{vec } \mathbf{Y}, (\text{vec } \mathbf{X}) \otimes \mathbf{I}_{rs}). \end{aligned}$$

Origin: Adapted from H Ex. 16.32(d)

**Solution.** Since

$$\text{vec } \mathbf{X} = \sum_{i,j} x_{ij}(\mathbf{e}_j \otimes \mathbf{e}_i), \quad \text{vec } \mathbf{Y} = \sum_{a,b} y_{ab}(\mathbf{e}_b \otimes \mathbf{e}_a),$$

the commutation action in the middle two factors gives

$$\begin{aligned}\mathbf{P}(\text{vec } \mathbf{X} \otimes \text{vec } \mathbf{Y}) &= \sum_{i,j,a,b} x_{ij} y_{ab} (\mathbf{e}_j \otimes \mathbf{e}_b \otimes \mathbf{e}_i \otimes \mathbf{e}_a) \\ &= \text{vec}(\mathbf{X} \otimes \mathbf{Y}).\end{aligned}$$

Therefore

$$\begin{aligned}d \text{vec } \mathbf{H} &= \mathbf{P}\{(d \text{vec } \mathbf{X}) \otimes \text{vec } \mathbf{Y} + \text{vec } \mathbf{X} \otimes (d \text{vec } \mathbf{Y})\} \\ &= \mathbf{P}(\mathbf{I}_{pq} \otimes \text{vec } \mathbf{Y}, (\text{vec } \mathbf{X}) \otimes \mathbf{I}_{rs}) d\mathbf{z}.\end{aligned}$$

Theorem 13.4 yields the Jacobian. ■

**Exercise 13.5** (Negative matrix powers). For nonsingular  $\mathbf{X} \in \mathbb{R}^{n \times n}$  and integer  $k \geq 1$ , prove

$$\begin{aligned} d(\mathbf{X}^{-k}) &= - \sum_{j=1}^k \mathbf{X}^{-j} (d\mathbf{X}) \mathbf{X}^{-(k+1-j)}, \\ D(\mathbf{X}^{-k}) &= - \sum_{j=1}^k (\mathbf{X}^{-(k+1-j)})' \otimes \mathbf{X}^{-j}. \end{aligned}$$

Origin: Course formulation from A&M Exs. 13.18, 13.21; H Eqs. (15.4.8), (15.8.15)

**Solution.** Apply (21) to  $(\mathbf{X}^{-1})^k$  and substitute  $d\mathbf{X}^{-1} = -\mathbf{X}^{-1}(d\mathbf{X})\mathbf{X}^{-1}$ . Roth's identity applied to each summand gives the Jacobian. ■

**Exercise 13.6** (Relative sensitivity of a linear solve). Let  $\mathbf{Ax} = \mathbf{b}$ , where  $\mathbf{A} \in \mathbb{R}^{n \times n}$  is nonsingular and  $\mathbf{b} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ . For simultaneous first-order changes, prove

$$\frac{\|\mathrm{d}\mathbf{x}\|_2}{\|\mathbf{x}\|_2} \leq \text{cond}_2(\mathbf{A}) \left( \frac{\|\mathrm{d}\mathbf{A}\|_2}{\|\mathbf{A}\|_2} + \frac{\|\mathrm{d}\mathbf{b}\|_2}{\|\mathbf{b}\|_2} \right), \quad \text{cond}_2(\mathbf{A}) = \|\mathbf{A}\|_2 \|\mathbf{A}^{-1}\|_2.$$

Origin: Course formulation from H Eq. (15.8.17)

**Solution.** Equation (26) gives

$$\|\mathrm{d}\mathbf{x}\|_2 \leq \|\mathbf{A}^{-1}\|_2 \{ \|\mathrm{d}\mathbf{b}\|_2 + \|\mathrm{d}\mathbf{A}\|_2 \|\mathbf{x}\|_2 \}.$$

Since  $\|\mathbf{b}\|_2 \leq \|\mathbf{A}\|_2 \|\mathbf{x}\|_2$ , division by  $\|\mathbf{x}\|_2$  proves the bound. ■

**Exercise 13.7** (An inverse bilinear form). For nonsingular  $\mathbf{X} \in \mathbb{R}^{n \times n}$  and constant  $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ , find the gradient of  $f(\mathbf{X}) = \mathbf{a}'\mathbf{X}^{-1}\mathbf{b}$ .

Origin: Adapted from A&M Exs. 13.22, 13.26; H Eqs. (15.8.15), (15.8.18)

**Solution.**

$$\begin{aligned}df &= -\mathbf{a}'\mathbf{X}^{-1}(\mathrm{d}\mathbf{X})\mathbf{X}^{-1}\mathbf{b} \\&= -\mathrm{tr}\{\mathbf{X}^{-1}\mathbf{b}\mathbf{a}'\mathbf{X}^{-1}\mathrm{d}\mathbf{X}\}.\end{aligned}$$

Hence

$$\nabla_{\mathbf{X}}f = -\mathbf{X}^{-T}\mathbf{a}\mathbf{b}'\mathbf{X}^{-T}.$$



**Exercise 13.8** (A Gram log-determinant). Let  $\mathbf{A} \in \mathbb{R}^{m \times m}$  satisfy  $\mathbf{A} = \mathbf{A}' \succ \mathbf{O}$ , and let  $\mathbf{X} \in \mathbb{R}^{m \times n}$  have full column rank. For

$$f(\mathbf{X}) = \log \det(\mathbf{X}'\mathbf{A}\mathbf{X}),$$

prove

$$\nabla_{\mathbf{X}} f = 2\mathbf{A}\mathbf{X}(\mathbf{X}'\mathbf{A}\mathbf{X})^{-1}.$$

Origin: Adapted from A&M Ex. 13.36(b); H Ex. 15.16; MC Ex. 20

**Solution.** Put  $\mathbf{S} = \mathbf{X}'\mathbf{A}\mathbf{X} \succ \mathbf{O}$ . Then

$$\begin{aligned} df &= \operatorname{tr}\{\mathbf{S}^{-1}((d\mathbf{X})'\mathbf{A}\mathbf{X} + \mathbf{X}'\mathbf{A}d\mathbf{X})\} \\ &= 2 \operatorname{tr}\{(\mathbf{A}\mathbf{X}\mathbf{S}^{-1})'d\mathbf{X}\}. \end{aligned}$$

Theorem 13.4 gives the result. ■

**Exercise 13.9** (Stationary set of the determinant). For  $f(\mathbf{X}) = \det \mathbf{X}$  on  $\mathbb{R}^{n \times n}$ , prove

$$\mathbf{X} \text{ is stationary} \iff \text{rank } \mathbf{X} \leq n - 2.$$

Origin: Adapted from A&M Ex. 13.33; H Eqs. (15.8.1)–(15.8.3)

**Solution.** By (27) and (41),

$$\mathbf{X} \text{ is stationary} \iff \text{adj } \mathbf{X} = \mathbf{O}.$$

The earlier adjugate-rank classification gives  $\text{adj } \mathbf{X} = \mathbf{O} \iff \text{rank } \mathbf{X} \leq n - 2$ . ■

**Exercise 13.10** (Kernel of the Gram differential). Let  $\mathbf{X} \in \mathbb{R}^{m \times n}$  have full column rank and  $\mathbf{G} = \mathbf{X}'\mathbf{X}$ . Characterize all  $\Delta$  satisfying

$$d\mathbf{G}_{\mathbf{X}}[\Delta] = \mathbf{O}.$$

For each  $\mathbf{S} = \mathbf{S}' \in \mathbb{R}^{n \times n}$ , construct an increment  $\Delta_{\mathbf{S}}$  satisfying  $d\mathbf{G}_{\mathbf{X}}[\Delta_{\mathbf{S}}] = \mathbf{S}$ . Deduce that the Jacobian of  $\mathbf{X} \mapsto \text{vech}(\mathbf{X}'\mathbf{X})$  has rank  $n(n+1)/2$ .

Origin: Course formulation from A&M Ex. 13.19(c); H Eqs. (15.4.3), (15.4.10), (15.5.3)

**Solution.** The equation is

$$\mathbf{X}'\Delta + \Delta'\mathbf{X} = \mathbf{0}.$$

Put  $\mathbf{K} = \mathbf{X}'\Delta$  and  $\mathbf{Z} = \Delta - \mathbf{XG}^{-1}\mathbf{K}$ . Then

$$\mathbf{K}' = -\mathbf{K}, \quad \mathbf{X}'\mathbf{Z} = \mathbf{0}, \quad \Delta = \mathbf{XG}^{-1}\mathbf{K} + \mathbf{Z}.$$

Conversely, every such skew-symmetric  $\mathbf{K}$  and  $\mathbf{Z}$  satisfy the equation. For  $\mathbf{S} = \mathbf{S}'$ , take

$$\Delta_{\mathbf{S}} = \frac{1}{2}\mathbf{XG}^{-1}\mathbf{S}; \quad \mathbf{X}'\Delta_{\mathbf{S}} = \Delta'_{\mathbf{S}}\mathbf{X} = \frac{1}{2}\mathbf{S}.$$

The kernel dimension is

$$\frac{n(n-1)}{2} + (m-n)n = mn - \frac{n(n+1)}{2}.$$

Rank-nullity gives rank  $n(n+1)/2$ , the dimension of the symmetric output. ■

**Exercise 13.11** (Inverse Jacobian as its own chain check). For nonsingular  $\mathbf{X} \in \mathbb{R}^{n \times n}$ , let  $\mathcal{I}(\mathbf{X}) = \mathbf{X}^{-1}$ . Verify directly that

$$D\mathcal{I}(\mathbf{X}^{-1})D\mathcal{I}(\mathbf{X}) = \mathbf{I}_{n^2}.$$

For nonsingular  $\mathbf{X} = \mathbf{X}'$ , verify the analogous identity in vech coordinates.

Origin: Adapted from A&M Ex. 13.38; H Eqs. (16.6.15), (16.6.19)

**Solution.** The unrestricted product is

$$\begin{aligned} & \{-\mathbf{X}' \otimes \mathbf{X}\} \{-\mathbf{X}^{-T} \otimes \mathbf{X}^{-1}\} \\ &= (\mathbf{X}'\mathbf{X}^{-T}) \otimes (\mathbf{X}\mathbf{X}^{-1}) = \mathbf{I}_{n^2}. \end{aligned}$$

For symmetric nonsingular  $\mathbf{X}$  and  $\mathbf{X}^{-1}$ , the earlier congruence- coordinate rule gives  $\mathbf{C}_\mathbf{A}\mathbf{C}_\mathbf{B} = \mathbf{C}_{\mathbf{AB}}$ . The two Jacobians are  $-\mathbf{C}_\mathbf{X}$  and  $-\mathbf{C}_{\mathbf{X}^{-1}}$ ; their product is  $\mathbf{C}_\mathbf{X}\mathbf{C}_{\mathbf{X}^{-1}} = \mathbf{C}_\mathbf{I} = \mathbf{I}_{n(n+1)/2}$ . ■

**Exercise 13.12** (Tangent blocks of a projector). Let  $\mathbf{P}(t) = \mathbf{P}(t)' = \mathbf{P}(t)^2$  be differentiable at 0, and put  $\mathbf{P} = \mathbf{P}(0)$  and  $\dot{\mathbf{P}} = \mathbf{P}'(0)$ . Prove

$$\begin{aligned}\dot{\mathbf{P}}' &= \dot{\mathbf{P}}, & \mathbf{P}\dot{\mathbf{P}}\mathbf{P} &= \mathbf{0}, \\ (\mathbf{I} - \mathbf{P})\dot{\mathbf{P}}(\mathbf{I} - \mathbf{P}) &= \mathbf{0}, & \operatorname{tr} \dot{\mathbf{P}} &= 0.\end{aligned}$$

Origin: Adapted from H Ex. 15.6 and Thm. 15.11.1; A&M Ex. 13.24

**Solution.** Differentiate  $\mathbf{P}' = \mathbf{P}$  and  $\mathbf{P}^2 = \mathbf{P}$ :

$$\dot{\mathbf{P}}' = \dot{\mathbf{P}}, \quad \dot{\mathbf{P}}\mathbf{P} + \mathbf{P}\dot{\mathbf{P}} = \dot{\mathbf{P}}.$$

Premultiplication and postmultiplication by  $\mathbf{P}$  gives  $2\mathbf{P}\dot{\mathbf{P}}\mathbf{P} = \mathbf{P}\dot{\mathbf{P}}\mathbf{P}$ . Replacing  $\mathbf{P}$  by  $\mathbf{I} - \mathbf{P}$  gives the second zero block. Taking traces in the differentiated idempotence identity gives  $2 \operatorname{tr}(\mathbf{P}\dot{\mathbf{P}}) = \operatorname{tr} \dot{\mathbf{P}}$ , while the first zero block and cyclicity give  $\operatorname{tr}(\mathbf{P}\dot{\mathbf{P}}) = 0$ . ■

**Exercise 13.13** (Null directions of the projector map). For the projector  $\mathbf{P}_{\mathbf{X}} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$  with full-column-rank  $\mathbf{X} \in \mathbb{R}^{n \times p}$ , prove

$$d\mathbf{P}_{\mathbf{X}}[\Delta] = \mathbf{0} \iff \Delta = \mathbf{X}\mathbf{C} \quad \text{for some } \mathbf{C} \in \mathbb{R}^{p \times p}.$$

Origin: Course formulation from H Thm. 15.11.1; A&M Ex. 13.24

**Solution.** If  $\Delta = \mathbf{X}\mathbf{C}$ , then  $\mathbf{M}\Delta = \mathbf{0}$ , so (36) vanishes. Conversely, multiply that equation on the left by  $\mathbf{M}$  and on the right by  $\mathbf{X}$ :

$$\mathbf{M}\{d\mathbf{P}_{\mathbf{X}}[\Delta]\}\mathbf{X} = \mathbf{M}\Delta\mathbf{R}\mathbf{X} = \mathbf{M}\Delta.$$

Thus zero differential implies  $\Delta = \mathbf{P}\Delta = \mathbf{X}(\mathbf{R}\Delta)$ . ■

**Exercise 13.14** (Spectral consistency of eigenvalue derivatives). Let  $\mathbf{A}(t) = \mathbf{A}(t)'$  have differentiable eigenpair branches  $(\lambda_i(t), \mathbf{q}_i(t))$ ,  $i = 1, \dots, n$ , at  $t = 0$ , such that

$$\mathbf{A}(t)\mathbf{q}_i(t) = \lambda_i(t)\mathbf{q}_i(t), \quad \mathbf{q}_i(t)'\mathbf{q}_i(t) = 1,$$

and  $\lambda_1(0), \dots, \lambda_n(0)$  are distinct. Prove

$$\sum_{i=1}^n \dot{\lambda}_i = \text{tr } \dot{\mathbf{A}}.$$

If  $\mathbf{A}$  is nonsingular, also prove

$$\left. \frac{d}{dt} \det \mathbf{A}(t) \right|_{t=0} = \sum_{i=1}^n \left( \prod_{j \neq i} \lambda_j \right) \mathbf{q}_i' \dot{\mathbf{A}} \mathbf{q}_i.$$

Origin: Course formulation from H Eq. (21.15.2); A&M Ex. 13.32

**Solution.** For the orthogonal  $\mathbf{Q} = (\mathbf{q}_1, \dots, \mathbf{q}_n)$ ,

$$\sum_i \dot{\lambda}_i = \sum_i \mathbf{q}_i' \dot{\mathbf{A}} \mathbf{q}_i = \text{tr}(\mathbf{Q}' \dot{\mathbf{A}} \mathbf{Q}) = \text{tr} \dot{\mathbf{A}}.$$

If  $\mathbf{A}$  is nonsingular,

$$\begin{aligned} \left. \frac{d}{dt} \det \mathbf{A}(t) \right|_0 &= (\det \mathbf{A}) \text{tr}(\mathbf{A}^{-1} \dot{\mathbf{A}}) \\ &= \left( \prod_j \lambda_j \right) \sum_i \lambda_i^{-1} \mathbf{q}_i' \dot{\mathbf{A}} \mathbf{q}_i, \end{aligned}$$

which is the displayed sum. 

**Exercise 13.15** (Splitting a repeated eigenvalue). Let

$$\mathbf{A} = \text{diag}(\lambda, \lambda, \mu), \quad \mathbf{E} = \begin{pmatrix} a & b & 0 \\ b & c & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda \neq \mu.$$

Find the two eigenvalues of  $\mathbf{A} + t\mathbf{E}$  that equal  $\lambda$  at  $t = 0$ . Show that their first-order rates are the two eigenvalues of  $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$ , and explain why a single formula  $\dot{\lambda} = \mathbf{q}'\mathbf{E}\mathbf{q}$  cannot define a unique gradient for an individually selected member of the repeated pair.

Origin: Course formulation from H §21.15, especially the simplicity hypothesis of Lemma 21.15.1

**Solution.** The leading block equals

$$\lambda \mathbf{I}_2 + t \begin{pmatrix} a & b \\ b & c \end{pmatrix}.$$

Hence its eigenvalues are

$$\lambda + t\theta_{\pm}, \quad \theta_{\pm} = \frac{a + c \pm \sqrt{(a - c)^2 + 4b^2}}{2}.$$

At  $t = 0$ , every unit vector  $\mathbf{q} = (q_1, q_2, 0)'$  is an eigenvector for  $\lambda$ , while  $\mathbf{q}'\mathbf{E}\mathbf{q}$  ranges over the Rayleigh quotients of the  $2 \times 2$  block. It is not a uniquely determined linear functional of  $\mathbf{E}$ . ■